



Transforming Dreams ! Redefining Future!

PLAYWRIGHT TEST AGENTS

Complete End-to-End Setup & Usage Handout

Prerequisites • Basic Tools • Playwright Project • GitHub Copilot + MCP • Agents • Usage

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Prereq		Node +		Playwright		Copilot		Init Agents
GitHub Acct	■	VS Code	■	Project	■	+ MCP	■	& Use

Who is this for? Absolute beginners. No prior knowledge of GitHub Copilot, MCP, or Playwright is required. Follow every step in order — install tools first, then plugins and agent setup.

Official Playwright Test Agents (Planner / Generator / Healer) + Playwright MCP for live browser control.

0. Prerequisites (Do This First)

Before installing any software or plugins, make sure you have a **GitHub account**. GitHub Copilot (required for Agent Mode and Playwright Agents) needs a GitHub login.

GitHub Account Required

- You need a GitHub username (login name) and password (or access via browser sign-in).
- If you do **not** already have a GitHub account, create one now at <https://github.com/signup>.
- Use a valid email address, choose a username, and set a password.
- Verify your email when GitHub sends the confirmation link.
- Keep your **username** and **password** (or sign-in method) ready — you will use them later when authorizing GitHub Copilot in VS Code.
- Free GitHub accounts are sufficient to start (Copilot has a free tier; paid plans give higher usage limits).

🔗 ■ You do **not** install Copilot yet. First we install the basic tools (Node.js, VS Code, Playwright). GitHub Copilot and MCP come in Step 3 after the project exists.

Other prerequisites checklist

- A computer with Windows, macOS, or Linux
- Internet connection (to download tools and browsers)
- About 2–3 GB free disk space (for Node.js, VS Code, and Playwright browsers)
- Admin rights to install software on your machine

1. Install Basic Tools (Node.js + VS Code + Playwright Extension)

Step 1.1 — Install Node.js (LTS)

1. Go to <https://nodejs.org> → click the big green **LTS** button → download and install.
2. Accept defaults and finish the installer.
3. Open a **new** terminal (or restart any open terminal) and verify:

```
node -v
npm -v
```

You should see version numbers (Node 20 or 22+ recommended). If you see a version → Success!

Step 1.2 — Install / Update Visual Studio Code (≥ 1.105)

1. Go to <https://code.visualstudio.com> → download and install the latest version.
2. Check version: open VS Code → Help → About (or Code → About on Mac).
3. You need **VS Code 1.105 or newer** for Playwright Agents to work properly.
4. If an update is available, install it and restart VS Code.

Step 1.3 — Install the Official Playwright Extension

1. In VS Code open the Extensions view (left sidebar icon or **Ctrl+Shift+X** / **Cmd+Shift+X**).
2. Search for **Playwright**.
3. Install **Playwright Test for VS Code** published by **Microsoft**.
4. After install you will see a Testing / Playwright sidebar.

🔗 ■ Basic tools are ready. Next we create a Playwright project (still no GitHub Copilot yet).

2. Create a Playwright Project

Step 2.1 — Initialize the project

1. Open VS Code.
2. Open a folder where you want the project (File → Open Folder) or create a new empty folder.
3. Open Terminal → New Terminal (or press **Ctrl+`**).
4. Run:

```
npm init playwright@latest
```

Answer the questions (recommended for beginners):

- Language: **TypeScript** (press Enter)
- Where to put your tests: press Enter (default is tests)
- Add a GitHub Actions workflow: **false** (or true if you want CI later)
- Install Playwright browsers: **true** (very important – press Enter)

Wait 1–3 minutes while Chromium, Firefox and WebKit download. When it finishes you have a working Playwright project with **playwright.config.ts** and sample tests.

🚀 ■ Project is ready. Now we install GitHub Copilot, enable Agent Mode, and add the Playwright MCP server (Step 3).

3. GitHub Copilot, Agent Mode & Playwright MCP Server

Now that Node.js, VS Code and the Playwright project exist, we install the AI plugins. You will use the GitHub account (username + password / sign-in) that you prepared in the Prerequisites.

Step 3.1 — Install GitHub Copilot Extensions

1. In VS Code open Extensions (**Ctrl+Shift+X** / **Cmd+Shift+X**).
2. Search for **GitHub Copilot** and install the official extension by GitHub.
3. Also install **GitHub Copilot Chat** (often installed together).
4. A Copilot icon appears in the status bar / activity bar.

Step 3.2 — Sign in with your GitHub Account

1. Click the Copilot icon or open Command Palette (**Ctrl+Shift+P**) → type **GitHub Copilot: Sign In**.
2. A browser window opens asking you to authorize Visual Studio Code / Copilot.
3. Sign in with the **GitHub username and password** (or existing session) you prepared in Prerequisites.
4. Click **Authorize** / **Continue**.
5. Return to VS Code – you should see “Signed in as <your-username>”.
6. If asked about free tier / Pro trial, you can start with the free tier.

If you do not have a GitHub account yet, go back to Prerequisites (page 2), create one at github.com/signup, then return here.

Step 3.3 — Enable Agent Mode

Agent Mode lets Copilot use tools (terminal, browser, MCP servers) instead of only answering questions.

1. Open Settings (**Ctrl+,**).
2. Search for: **chat.agent.enabled**
3. Enable the checkbox **Chat: Agent Enabled**.
4. Open Chat (Command Palette → “Chat: Open Chat”) and in the dropdown choose **Agent** (not only Chat).

You should now see an **Agent** option in the Copilot Chat panel.

Step 3.4 — Install Playwright MCP Server

The MCP server lets the agent open a real browser, read the page, click and type. Without it the agents mostly generate code from memory.

Option A – One-liner (easiest)

Open a terminal **outside** VS Code (or external PowerShell) and run:

```
code --add-mcp '{"name":"playwright","command":"npx","args":["@playwright/mcp@latest"]}'
```

Then restart VS Code.

Option B – Project file

In your project create folder **.vscode** and file **mcp.json**:

```
{
  "servers": {
    "playwright": {
      "command": "npx",
      "args": ["@playwright/mcp@latest"]
    }
  }
}
```

Click the play button above the server name, or use Command Palette → **MCP: List Servers** → start Playwright.

Option C – Marketplace

Extensions → search **@mcp playwright** → Install → trust the server when asked.

🔗■ After MCP is set up: open Chat → Agent mode → Tools / Configure Tools → enable Playwright tools (browser_navigate, browser_click, browser_snapshot, etc.).

4. Initialize Playwright Test Agents & Seed File

Step 4.1 — Run init-agents

In the same project terminal (inside your Playwright folder) run:

```
npx playwright init-agents --loop=vscode
```

This creates:

- Agent definition files (under .github/ or chat modes)
- A starter **tests/seed.spec.ts** used by the agents as a launch pad
- Ready-to-use Planner, Generator and Healer agents for VS Code

After the command finishes: Completely close and reopen VS Code, then open your project folder again.

Step 4.2 — Edit the Seed File

Open **tests/seed.spec.ts** and point it at your app (or the public demo):

```
import { test, expect } from '@playwright/test';

test('seed', async ({ page }) => {
  await page.goto('https://demo.playwright.dev/todomvc'); // ← change to your URL
  // optional: accept cookies, login, etc.
});
```

The agents run this seed first so they start from a known good state of your application.

Step 4.3 — Verify setup

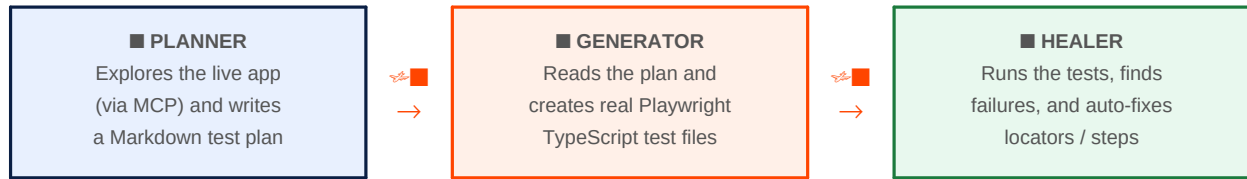
1. Open Copilot Chat → switch to **Agent** mode.
2. Confirm Playwright MCP tools are enabled (Tools button).
3. Type:

Is this the Playwright Planner agent?

If you get a confirmation that the Playwright Planner (or related agent) is active → setup is successful!

5. How Playwright Agents Work (The Agentic Loop)

Three specialized agents work together. You speak in plain English; they explore, write code, and fix failures.



Typical project structure after agents run:

- **.github/** or agent definitions – instructions + tools for the three agents
- **specs/** – human-readable Markdown test plans from Planner
- **tests/** – generated Playwright test files + seed.spec.ts
- **playwright.config.ts** – normal Playwright configuration

6. Sample Prompts – Ready to Copy & Paste

Open Chat → choose **Agent** mode → paste one of the prompts below. Mention the seed file or the plan file so the agent knows the context.

■ **PLANNER – Create a Test Plan**

Using the seed test (tests/seed.spec.ts), explore the application and create a detailed Markdown test plan for the main user flows of the TodoMVC app. Cover adding, completing, editing and deleting todos. Save the plan under specs/todo-basic-operations.md

Generate a test plan for the guest checkout flow on https://your-app.com. Include the happy path and at least three negative scenarios (empty cart, invalid email, payment failure). Use the seed.spec.ts as the starting point.

■ **GENERATOR – Turn Plan into Real Tests**

Take the test plan in specs/todo-basic-operations.md and generate executable Playwright TypeScript tests. Prefer getByRole and getByTestId locators. Put each scenario in its own file under tests/.

Generate Playwright tests for the checkout scenarios described in the plan. Include assertions for success messages, URL changes and cart count. Use the seed test as the base.

■ **HEALER – Debug & Auto-Fix Failing Tests**

Run the test suite and heal any failing tests. Focus on the tests related to adding a todo. If a locator is broken, update it to a more resilient one and re-run until it passes.

The test tests/create/add-valid-todo.spec.ts is failing. Replay the steps, inspect the current UI with the browser tools, suggest a fix and apply the patch. Re-run the test to confirm it passes.

Heal all currently failing tests in the project. For each failure explain what changed and what you fixed.

7. Running Tests & Classic Debugging in VS Code

Even with agents you still use the normal Playwright tools:

- Testing sidebar (beaker icon) – click the play button next to any test or the whole suite.
- Debug: set a breakpoint → right-click the test → Debug Test.
- Enable “Show browser” in the Playwright sidebar to watch the live browser.
- After a run click the Trace icon for time-travel debugging.
- When a test fails you can also click the sparkle / Fix with AI icon for a Copilot suggestion.

Useful terminal commands:

```
npx playwright test # run all tests
npx playwright test --headed # see the browser
npx playwright test --debug # step-by-step debug
npx playwright show-report # open HTML report
npx playwright test --ui # interactive UI mode
```

8. Beginner Tips & Common Gotchas

- Have your GitHub username and password ready before Step 3 (sign-in to Copilot).
- Always keep **seed.spec.ts** working first. If the seed cannot open your app, the agents cannot explore either.
- After updating Playwright, re-run **npx playwright init-agents --loop=vscode** so agent definitions stay current.
- Make sure you are in **Agent** mode (not plain Chat) and that Playwright MCP tools are enabled.
- Prefer role-based locators (`getByRole`) – the agents already try to generate them.
- Start with the public demo <https://demo.playwright.dev/todomvc> before pointing agents at a private application.
- The Healer is powerful but not perfect – always review the changes it makes before committing.
- If MCP tools do not appear: Command Palette → MCP: List Servers → start Playwright, or restart VS Code.
- Free Copilot tier has usage limits; heavy agent use may require a paid plan.

🚩 Full End-to-End Journey Recap

0. Create/prepare GitHub account (username + password) → 1. Install Node.js + VS Code + Playwright extension → 2. `npm init playwright@latest` → 3. Install Copilot, sign in with GitHub, enable Agent Mode, install Playwright MCP → 4. `npx playwright init-agents --loop=vscode` → Edit seed → Talk to Planner → Generator → Healer → Run & enjoy!

Quality Thought

Transforming Dreams! Redefining Future!

This handout is designed for absolute beginners. Install tools first, then plugins. Follow every step in order and you will have a complete AI-powered Playwright Agents environment.

Official docs: <https://playwright.dev/docs/test-agents>

Playwright MCP: <https://playwright.dev/docs/getting-started-mcp>

VS Code Agents & MCP: <https://code.visualstudio.com/docs/copilot/customization/mcp-servers>